

A Hybrid Physics-Based Data-Driven Framework for Anomaly Detection in Industrial Control Systems

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Abstract—A method referred to as PbNN is proposed to detect cyber-physical attacks through the identification of resulting anomalies in the process dynamics of the underlying ICS. Unlike existing anomaly detectors based on an abstract knowledge acquired from operational data, PbNN utilizes the design knowledge of ICS to learn the complex relationships among the correlated components. Such relationships are accurately modeled using operational data through the application of the deep convolution neural network. The proposed detector was implemented and evaluated in an operational secure water treatment plant by launching several real-time stealthy and coordinated attacks. The results indicate that PbNN outperforms the existing state-of-the-art machine learning anomaly detectors when compared using detection accuracy and the rate of false alarms.

Index Terms—Anomaly detection, critical infrastructure (CI), deep convolution neural networks, industrial control systems (ICSs), stealthy attacks, water treatment plant.

I. INTRODUCTION

INDUSTRIAL control systems (ICS) are found in critical infrastructure (CI), such as a water treatment plants, oil refineries, and nuclear power plants. An ICS integrates physical and cyber components to control the underlying process in such plants. In this work, the problem of detecting anomalies in a process under the control of an ICS is addressed.

Problem Context: The physical process in a plant¹ is considered to be in its normal operating state when it performs in accordance with its design. However, faults in the physical components or cyber attacks may lead to process anomalies, i.e., process behavior deviating from the normal. The problem of detecting such anomalies is addressed in this

work. The anomaly detection problem solved in this addressed work problem is to create a method to detect and report process anomalies resulting from a cyber attack. An attack could occur when a skilled attacker successfully bypasses the existing defense mechanisms to gain unauthorized access for compromising: 1) sensors, e.g., manipulating the sensor measurements sent to the PLCs; 2) actuators, e.g., altering the control signals received from the PLCs; and 3) the PLCs, e.g., mutating its firmware or control codes. In this article, we propose a physics-based neural network (PbNN) that monitors the process in real time to detect a wide range of process anomalies.

Solution Strategy: A cyber attack against ICS could be defined as “intentional actions trying to cause undesired effects in the physical world” [1]. A formal model representing the behavior and interrelationship among the state variables² is built and any perceived deviation between the actual and predicted behavior leads to an alert. PbNN identifies such abnormal deviations in operational data from the plant. PbNN operates in two phases. In the “offline phase,” the interactions among the state variables are modeled using the deep convolutional neural network (DCNN). In the “online phase,” anomalies are detected by comparing the expected behavior of the plant obtained from DCNN against the observed behavior.

Novelty of the Proposed Approach: Significant research, focusing on the exploitation of techniques from machine learning for detecting process anomalies, has been reported [2]–[5]. However, to the best of our knowledge, the methods reported are physics unaware and directly apply deep-learning algorithms on the operational data. As an ICS consists of heterogeneous state variables, i.e., continuous and discrete valued, these approaches may fail to capture the hidden inherent correlations due to the omission of design knowledge. Furthermore, these approaches compare directly the actual and predicted behavior in terms of the magnitude of the state variables, which makes them less effective in real-time detection of indirect³ and multipoint coordinated attacks⁴ that mask the impact of an attack on multiple state variables.

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¹The term “plant” is used here to refer to a specific system and includes the ICS and the controlled physical process.

²A state variable measures one attribute of one component of the plant, e.g., water level in a tank.

³An indirect attack is a targeted attack on a one state variable but also affects the behavior of another variable.

⁴A multipoint coordinated attack targets more than one state variables to deceive any attack detection mechanisms.

Contrasted with the existing methods, PbNN automatically learns the essence of the physical process controlled by an ICS using the interactions among the set of correlated state variables obtained from plant design. Such relationships are effectively modeled through the application of DCNN. Anomaly detection is carried out by quantifying the nonrandomness of the residual sequence generated through the actual and predicted values in a live plant. The novelty and generality of the methodology used in PbNN are a key advancement in the design of secure ICS. Briefly, the novelty of PbNN lies in the design and evaluation of a systematic procedure: 1) for building the physical model based on the plant design and operational dataset that integrates the discrete and continuous state variables and 2) its evaluation in a live plant by launching a range of single and multipoint attacks. From experience, training and evaluating an anomaly detector on data collected from a plant, i.e., nonlive evaluation, as opposed to doing so in a live plant, lead to different results. Such results often favor the detection method whereas when actually applied in a live plant, the performance turns out to be radically different from that found during a nonlive evaluation.

Research Questions: *RQ 1:* How accurately PbNN learns process dynamics of the underlying ICS through the application of design knowledge and operational data? *RQ 2:* How effective is PbNN in detecting process anomalies compared with the existing state-of-the-art machine learning-based anomaly detectors? *RQ 3:* Does PbNN support the plant operator in the identification of attack target(s)?

Contribution:

- 1) PbNN, a physics-based neural network anomaly detection framework is proposed for distributed ICS that operates over the interactions among the physical components.
- 2) The automated learning strategy in PbNN can effectively learn the process dynamics and control flow through the application of design knowledge and operational data set. This allows PbNN to comprehensively simulate the behavior of individual components, which is utilized for the detection of potential anomalies.
- 3) The proposed solution does not require any changes to the hardware or software of the ICS components and independent of the data source. It can operate in a distributed manner for the anomaly detection process without altering the data flow rate.
- 4) The performance of PbNN was evaluated on the realistic secure water treatment (SWaT) plant by launching several real-time cyber-physical attacks. Furthermore, the performance is also compared against the existing state-of-the-art anomaly detectors designed for SWaT.

Organization: The remainder of this work is organized as follows. Section II is a review of recent research in design and data-centric approaches for detecting process anomalies in ICS. Assumptions and system model used in this work are in Section III. Section IV provides a detailed description of the modules in PbNN. Section V presents a case study to evaluate the effectiveness of the proposed approach followed by details of the experiments conducted. Performance analysis is presented in Section VI. Section VII offers conclusions of

this work with insights gained and proposes future directions for research in anomaly detection in the context of ICS.

II. RELATED WORK

Umer *et al.* [6] proposed DeC, a data-centric approach-based rule generation from the operational data using an associative rule mining (ARM) algorithm. Initially, using a feature selection algorithm, the redundant and least significant features are identified and removed. Subsequently, using ARM, frequent itemsets from the data subset are generated. Finally, the itemsets are converted to rules for process monitoring. These rules and their effectiveness in detecting process anomalies were experimentally compared against the invariants generated by DAD [7] on the SWaT dataset. As an ICS consists of a large number of sensors and actuators, the generation of rules using ARM is impractical for legacy plants. Also, neglecting few state variables from the rule generation process minimizes the detection coverage of the anomaly detector. In contrast, PbNN automatically identifies the correlated components using the plant's design as expressed in its P&ID and their relationship is accurately modeled using DCNN in a minimal time.

Hu *et al.* [8] presented a hybrid framework by integrating the Kalman filter and permutation entropy approach for anomaly detection in ICS. As an extension, the Kalman filter was integrated with residual skewness by Hu *et al.* [9] for the same research problem. The performance of both approaches was evaluated in a simulated environment by launching single point stealthy attacks. In PbNN, we consider both single and multipoint coordinated stealthy attacks launched against an operational plant.

Raman *et al.* [10] proposed a deep autoencoder-based anomaly detector for operational SWaT. In this work, a comparative study was performed by varying the amount of design knowledge in building the deep autoencoder (DAE) for effective detection of process anomalies. Three variants of the DAE architecture were built with varied design knowledge and one with "stagewise grouping" was found to have a better performance than others. One of the major limitations that degrades the performance of DAE was the way in which the relationship between the sensor and actuators is modeled. Since they considered only the spatial dependencies across the components, the data from all state variables are scaled down to a uniform range for effective modeling. By doing this, the higher valued state variables, such as level sensors and chemical sensors, dominate the lower valued variables, such as pumps, valve, and flow meter sensors. PbNN is independent of the operating range of components since it operates over the spatiotemporal dependency.

Kravchik and Shabtai [4] proposed a lightweight neural network anomaly detection framework using 1-D CNN and autoencoder. In this work, the neural network architectures are trained with the time and frequency domain of data collected from the operational plant for anomaly detection. To overcome the computational complexity due to the existence of high-dimensional data, principal component analysis (PCA) was utilized. This work was found to be an extension of [3]

with a primary focus on evaluating the robustness of the neural network architecture to adversarial machine learning attacks. The experimental evaluation was carried out using three public datasets, namely, the SWaT dataset, BATADAL, and WADI. PbNN does not require any dimensionality reduction technique since the set of state variables with the least dependency is neglected at the initial stage using the design knowledge. Furthermore, instead of direct application of the neural network architecture on the entire dataset, PbNN operates in a distributed manner.

Raman *et al.* [11] proposed an unsupervised approach for detecting cyber attacks in water treatment plants. In this work, a multilayer perceptron (MLP) neural network was utilized as a time-series predictor to predict the sensor measurements using temporal dependencies. Furthermore, through the application of a statistical technique, the actual and predicted measurement was compared to detect anomalies. Similar studies are available in [5] and [12]–[15] were several deep learning algorithms such as long short-term memory (LSTM), recurrent neural network (RNN), generative adversarial neural (GAN), etc., and are integrated with statistical techniques for the anomaly detection process. The effectiveness of these approaches was experimentally evaluated using several ICS datasets, such as SWaT, WADI, and Tennessee Eastman process (TES). A notable similarity in these approaches is the anomaly scoring criteria. A safety limit to which the deviation between the actual and predicted value can exist is based on the magnitude of the residual sequence. Such an approach is not practically suitable for detecting stealthy attacks. Furthermore, these approaches are based on temporal dependencies, thereby limiting their applicability to continuous state variables. In contrast, PbNN operates over the distribution and nonrandomness contained in the residual sequence for effective detection of both stealthy and indirect attacks. Also, PbNN is suitable for monitoring the behavior of both discrete and continuous time state variables.

III. MATHEMATICAL FORMULATION AND ASSUMPTIONS

A mathematical formulation of the anomaly detection problem in ICS and the assumptions made in this work are described next.

A. Mathematical Formulation

Let $X(t)$ be an n -dimensional state vector representing the measured values of state variables of the plant at time t . We write $X(t) = X_c(t) \cup X_d(t)$, where $X_c(t)$ and $X_d(t)$ are the continuous and discrete state variables, respectively. For example, in SWaT [16], the state variable denoting the state of pump P101 in stage P1 is a discrete state variable while the water level in tank T101, measured by sensor LIT101, is a continuous state variable.

The construction and use of the anomaly detector proceed in the following steps. First, the set of interdependent state variables is identified from the plant design. Next, the nonlinear relationships among the identified state variables are modeled and the behavior of the system ($\hat{x}(t)$) is forecast. Finally, the quality of the estimate, i.e., residue ($r(t) = x(t) - \hat{x}(t)$),

is determined by comparing the measured outcome ($x(t)$) of the system with $\hat{x}(t)$ for the detection of anomalies, i.e., outliers. A statistical model $P_r(t)$ is postulated for the distribution of the residual sequence during normal operating conditions. Anomalies are defined as the residues whose probability is greater than a predefined threshold under this model. The statistical hypotheses tested are [17]:

H_0 : r is drawn from $P_r(t)$, and H_1 : r is drawn from noninformative distribution.

According to the minimum probability of error rule for this test, the null hypothesis H_0 is rejected when $P_r(t) > \tau$, where τ is the constant of noninformative distribution.

B. Models and Assumptions

1) *Attacker Model*: A cyber attack against an ICS is a sequence of actions initiated by the attacker against one or a group of physical or cyber components to realize the intended objective. We assume that the attacker's objective is to alter the physical process while remaining undetected. Thus, passive attacks that do not affect the process flow are excluded from this work. We assume an insider threat to the plant implying that the adversary has access to the design specification and dimensions of the components of the plant needed to launch the attack. Furthermore, the adversary can inject false information closer to the normal behavior of the system to reduce the chances of the plant operator noticing suspicious behavior.

The adversary can compromise a subset of the PLCs to launch coordinated attacks and can have complete control over the compromised PLCs, i.e., its control logic can be accessed and modified in real time. The values of state variables in the compromised PLCs can be spoofed during the plant operation and recorded in the data historian. Finally, the ICS has built-in safety mechanisms, like a circuit breaker, mechanical switches, and alarm generator should not be activated by the attacker's action.

2) *PbNN*: We assume that the system in which our proposed anomaly detector is deployed is robust and not compromised. It is operated in a system that is isolated from the network through which the data transfer occurs among the plant components. Thus, we assume that a cyber attack will not be able to compromise both the ICS and the anomaly detector. Furthermore, we also assume that there exists one-way communication between the data historian and detector. This implies that the detector can extract the values of state variables from the data historian during plant operation but cannot alter or send back any data into the plant network.

IV. PBNN: ARCHITECTURE AND BUILDING BLOCKS

The overall architecture shown in Fig. 1 consists of several modules of the proposed deep learning-based anomaly detection framework. The operation of PbNN takes place in two phases, namely: 1) an offline phase for learning the interactions among the state variables using design knowledge and operational dataset and 2) an online phase for real-time monitoring of the process plant to detect the existence of anomalies. A detailed explanation of each module in PbNN is as follows.

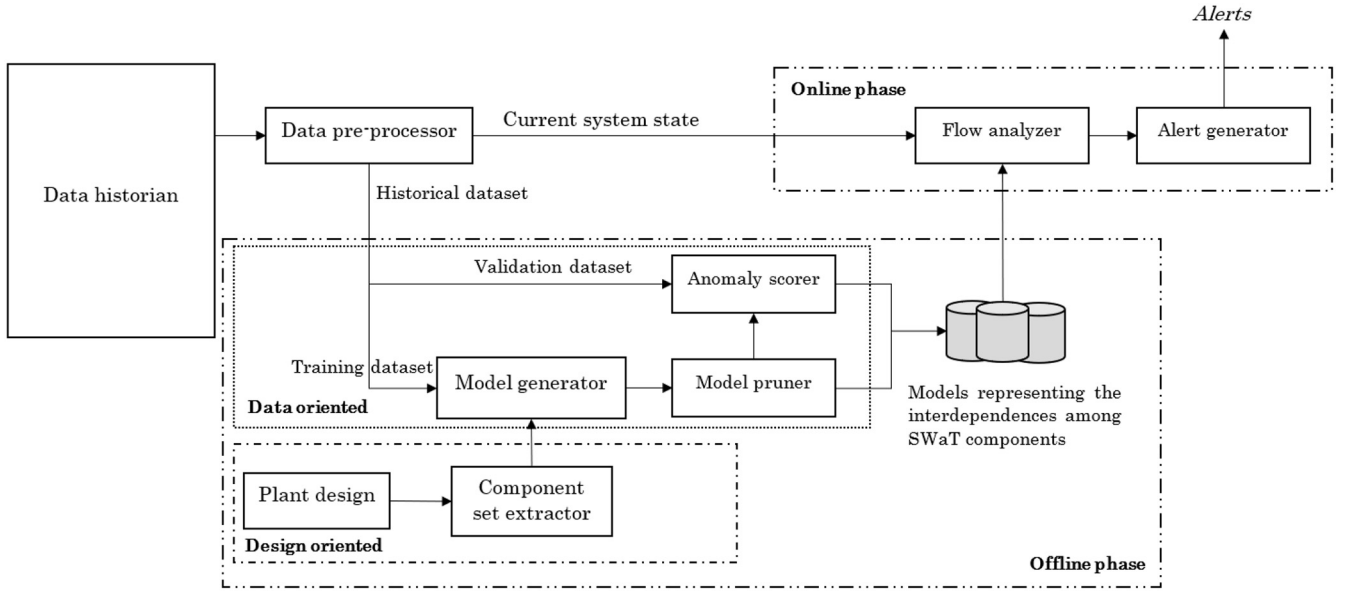


Fig. 1. PbNN: Architecture of proposed real time anomaly detection framework for ICS.

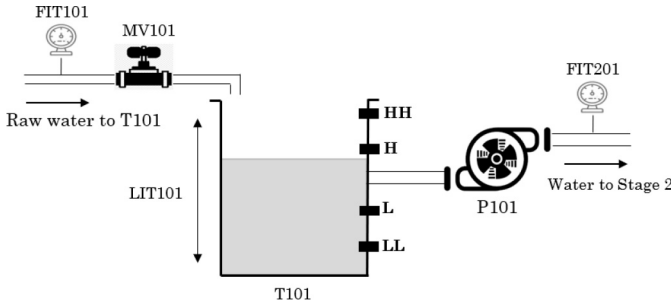


Fig. 2. Components relevant to this work in stage P1 of SWaT. Markers LL, L, H, and HH are used by the PLC to control MV101 and P101. These markers are also used in the design of attacks listed in Tables I and II.

A. Component Set Extractor

The component set extractor uses a design-oriented approach that provides a set of interdependent state variables obtained from the plant design. In this work, we rely on the process invariants to learn the interactions among the state variables. A “process invariant”, hereafter referred to as “invariant,” is a mathematical relationship among the physical properties of the process. As there exist several types of invariants discussed in [7] and [18], this work deals with invariants to estimate a state variable via that of its associated components controlled by a PLC.

To illustrate, consider stage P1 of SWaT as in Fig. 2. Here, we have the set of state variables, $n = \{n_1 : MV101, n_2 : LIT101, n_3 : P101, n_4 : FIT101, n_5 : FIT201\}$. State variables n_1 and n_3 are discrete while the remaining are continuous. A possible representation of the state space of n follows.

$$\begin{aligned} V_1 &= \{\text{Open, Closed}\} \\ V_2 &= \{x_1 \in \mathbb{R}\} \\ V_3 &= \{\text{ON, OFF}\} \end{aligned}$$

$$\begin{aligned} V_4 &= \{x_2 \in \mathbb{R}\} \\ V_5 &= \{x_3 \in \mathbb{R}\}. \end{aligned}$$

At time t , let $x_1(t)$ be the water level in T101 and $x_2(t)$ and $x_3(t)$ denote, respectively, the rates of inflow and outflow. An invariant to estimate the water level in T101 at time step $t + 1$ is defined as

$$x_1(t + 1) - x_1(t) = \delta(x_2(t) - x_3(t)) \quad (1)$$

where δ is a constant computed based on the physical properties of the tank and units of the flow rate. Additional invariants exist for each state variable in SWaT. Rearranging (1), we get

$$x_1(t + 1) = x_1(t) + \delta(x_2(t) - x_3(t)). \quad (2)$$

Equation (2) is valid for an ideal sensor. Furthermore, practically deriving the relationship among the state variables for a given invariant is a complex task due to the existence of higher order and nonlinear correlations. To overcome this complexity, we use a data-centric approach resulting in the physics-based neural network model.

B. Model Generator and Model Pruner

Using (2), let us define a nonlinear function $f()$ given by

$$x_1(t + 1) = f(x_1(t), x_2(t), x_3(t)). \quad (3)$$

From (3), for a given plant with normal behavior of components and its corresponding invariants, we can build $f()$ that can estimate the future behavior of components with minimal error. The “model generator” module builds a physical model that represents the interactions among the state variables through the application of DCNN. From the operational dataset, the values of the independent state variable ($x_1(t), x_2(t), x_3(t)$) for a given invariant are provided as input to the DCNN architecture and trained in such a way that the exact or closer values of the dependent state variable ($x_1(t+1)$) is obtained at the output layer. The outstanding performance

of DCNN variants in addressing several regression problems is not only due to the careful design of the corresponding CNN architecture but also the proper selection of hyperparameters. They are fine-tuned with a constraint to minimize the prediction error as in

$$\text{minimize } f_e = \frac{1}{n} \sum_{i=1}^n (x_i(t) - \hat{x}_i(t))^2; \text{ w.r.t. } H_p \quad (4)$$

where f_e is the prediction error, $x_i(t)$ and $\hat{x}_i(t)$ are, respectively, the actual and predicted states of the i th component at the t -th timestep, and H_p is the set of hyperparameters. The problem of fine-tuning the hyperparameters in a CNN can be viewed as multimodal function optimization as the hyperparameters include parameters in convolution layer (number of filters, kernel size, and padding), pooling layer (pooling type and padding), and fully connected layer (number of hidden layers, number of hidden neurons, and activation function). Manual tuning of hyperparameters is a costly trial-and-error method. Furthermore, the heterogeneous nature and the volume of multivariate data with nonlinear relationships generated by an ICS make manual tuning difficult.

Hence, in this work, we adopt an automated grid search for fine-tuning the hyperparameters of DCNN, where multiple DCNN models with different values of hyperparameters are generated by the model generator module. The model pruner will select the best model on the basis of Theil's U1 (T_{u1}) statistical score defined in

$$T_{U1} = \frac{\left[\sum_{i=1}^n (x_i - \tilde{x}_i)^2 \right]^{\frac{1}{2}}}{\left[\sum_{i=1}^n (x_i)^2 \right]^{\frac{1}{2}}}. \quad (5)$$

The value of T_{U1} will always be in the interval of $[0,1]$, where 0 indicates that x and $\tilde{x}$ are same.

C. Anomaly Scorer

The anomaly scorer module computes the acceptable range to which the predicted state of a component can deviate from the actual state. Once the best DCNN model is obtained for a given invariant through model pruner, its predicted state of the corresponding component is compared against its observed state for anomaly detection. In order to ensure the normal operation of the plant, the computed residue $r(t)$ must be within an acceptable range (τ) as discussed in Section III-A. Due to several factors, such as nonlinearity, aging of components, misconfiguration, temporal glitches, etc., there exists a significant challenge in computing the optimal value of τ . Moreover, the anomaly detection process is sensitive to these parameters and a poor choice will likely lead to an unacceptable number of false alarms.

Generally, the residual series generated during the normal operation of a plant is random. During a cyber attack, there exists some sort of nonrandom dynamics in the series. Using statistical approaches, these patterns, or regularities, can be characterized for the effective detection of process anomalies. In the reported study, CUMulative Sum (CUSUM) approach is integrated with DCNN to monitor and evaluate the magnitude of residues. CUSUM is a commonly used stateful approach for detecting abnormal deviations that correspond to process

anomalies. It computes the CUSUM of the residual sequence to detect minor deviations in their magnitude. In order to minimize the false positives, we calculate both positive and negative side deviation of the residues using (6) and (7), respectively [19]

$$P(t) = \max(0, r(t) - \text{target} - b) \quad (6)$$

$$N(t) = \min(0, r(t) - \text{target} + b) \quad (7)$$

where $P(t)$ and $N(t)$ are, respectively, the high and low CUSUMs, target represents the safety limit, and b is the allowable slack. Furthermore, we introduce two parameters, namely, upper control limit (UCL) and lower control limit (LCL), which are computed empirically using the values of P and N . During the testing phase, when $P(t) > \text{UCL}$ or $N(t) < \text{LCL}$, the behavior of the component is considered suspicious.

D. Flow Analyzer and Alert Generator

Flow analyzer and alert generator constitute the online phase of PbNN. They act as an interface between the overall system and the plant operator. They continuously monitor and analyze the operation of each subprocess through the knowledge obtained from the offline phase to detect anomalies.

These modules receive the current measurements for all state variables and perform a consistency check analysis through the physical models obtained from the model pruner. An alarm is raised by the alert generator when an anomaly is detected. In practice, there may exist several invariants to estimate the value of a state variable. For example, as discussed in Section IV-A, the value of LIT101 can also be estimated through the status of n_1 , n_3 , and n_4 . As the several invariants for estimating the values of individual state variables are converted to physical models, initially they are consolidated by the flow analyzer. Furthermore, through the parallel execution process, the flow analyzer identifies which model of the physical system deviated from the expected. Through this process, the proposed anomaly detector not only detects an anomaly but also localizes it for forensics.

V. CASE STUDY

The effectiveness of the proposed anomaly detector was assessed experimentally through a case study by designing stealthy attacks against SWaT, launching them, and observing the consequent plant behavior. In this section, we discuss the attack design and its launch procedure used in the case study.

A. Attack Design

An attack model is defined as a three tuple (A_T, A_P, A_G) , where A_T corresponds to the type of attack launched through a finite set of points (A_P) with an objective (A_G). An objective (A_G) of the attacker can be represented as a statement. For example, "damage the motorized valve (MV101)," or "alter the pH of water from a distribution plant." Furthermore, an attack is considered to be *single point*, when only one component, e.g., a sensor, serves as its target, i.e., $A_P = 1$. As there exist several attack types specific to the plant discussed in [20], we

TABLE I
SINGLE POINT STEALTHY ATTACKS LAUNCHED AGAINST SWAT TESTBED ($X = [3-5 \text{ mm}]$; $Y = [0.1-0.3 \text{ cm/hr}]$)

Attack ID	Targeted sensor	Intention	Launch procedure
A1	LIT101	Overflow tank T101	Adversary gains control of point C in Figure 3; Whenever LIT101 is greater than 750 mm, the value is reduced by X mm and sent to PLC1; PLC1 does not CLOSE MV101 since LIT101 is less than H.
A2		Underflow tank T101	Adversary gains control of point C in Figure 3. Whenever LIT101 is less than 550 mm, the value is increased by X mm and send it to PLC1; PLC1 does not OPEN MV101 since LIT101 is greater than L.
A3	FIT101	Slow filling of T101	Adversary gains control of point A in Figure 3. Whenever FIT101 is greater than 2.5 mm/hr, the value is reduced by Y cm/hr for every second and send it to PLC1
A4	LIT301	Overflow tank T301	Adversary gains control of point F in Figure 3; Whenever LIT301 is greater than 950 mm, the value is reduced by X mm and send it to PLC3; PLC3 does not instruct PLC2 to CLOSE MV201 since LIT301 is less than H
A5		Underflow tank T301	Adversary gains control of point F in Figure 3; Whenever LIT301 is less than 850 mm, the value is reduced by X mm and send it to PLC3; PLC3 does not instruct PLC2 to OPEN MV201 since LIT301 is greater than L
A6	FIT201	Disturb chemical dosing	Adversary gains control of point E in Figure 3; Whenever FIT201 is greater than 1.5 mm/hr, the value is reduced by Y cm/hr for every second and send it to PLC2

TABLE II
MULTIPOINT COORDINATED STEALTHY ATTACKS LAUNCHED AGAINST SWAT TESTBED ($X = [3-5 \text{ mm}]$; $Y = [0.1-0.3 \text{ cm/hr}]$)

Attack ID	Targeted sensors	Intention	Launch procedure
A7	LIT101 & FIT101	Overflow tank T101	Adversary gains control of point C and A in Figure 3; Whenever LIT101 is greater than 750 mm, both LIT101 and FIT101 are reduced by X mm and Y cm/hr for every second respectively.
A8		Underflow tank T101	Adversary gains control of point C and A in Figure 3; Whenever LIT101 is less than 550 mm, both LIT101 and FIT101 are increased by X mm and Y cm/hr for every second respectively.
A9	LIT301 & FIT201	Overflow tank T101	Adversary gains control of point F and E in Figure 3; Whenever LIT301 is greater than 950 mm, both LIT301 and FIT201 are reduced by X mm and Y cm/hr for every second respectively
A10		Underflow tank T101	Adversary gains control of point F and E in Figure 3; Whenever LIT301 is less than 850 mm, both LIT301 and FIT201 are increased by X mm and Y cm/hr for every second respectively

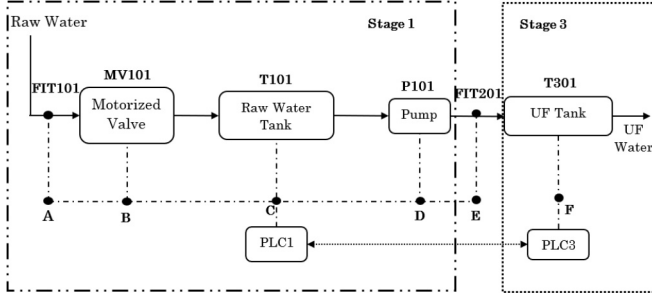


Fig. 3. SWaT components involved in the attack and detection scenarios.

have considered stages P1-P3 in our case study. The stealthy attacks, each with a specific intention, are launched against the components in the selected stages as shown in Fig. 3. The attacks considered in this case study are summarized in Tables I and II with their respective intentions.

B. Attack Launch

Prior to the launch of stealthy attacks, the plant was operated and brought to a steady state. No attacks were launched when the plant was in a transient state, i.e., during startup and shut-down. In order to ensure the plant is operating in the steady state, we monitor the quantity of filtered water produced at different time instances. Initially, the plant was operated in a normal state for more than an hour and then a series of attacks is launched. To avoid the cascading effects of one attack on the next, a gap of 10 min was maintained between the successive attacks. Furthermore, each attack was launched five times in various steady states of SWaT to analyze the response of

the anomaly detectors. The general procedure followed during the launch of stealthy attacks is as follows.

- 1) Identify the memory location (tag) in the PLC, where the measurement of the targeted sensor will be saved.
- 2) Compromise the wireless link between the PLC and the SCADA workstation.
- 3) Change the tag to a different value, so that the PLCs assume the manipulated value to be a true state of the component, which leads to incorrect control actions.

VI. EVALUATION

In this section, we demonstrate the performance of PbNN over the existing state-of-the-art machine learning-based anomaly detection approaches through a set of attacks discussed in Section V.

A. Experimental Setup

The proposed anomaly detector was designed using Python 3.7 and the Keras library on a system with Intel Core i7 Processor 2.50 GHz, 16 GB of RAM running Windows 10 OS. For training, i.e., modeling the invariants, and validation, i.e., computing the parameters discussed in Section IV-C, we utilized the records corresponding to the normal operation of the plant obtained from the SWaT dataset. Furthermore, the anomaly detector was tested on data received in real time from the operational SWaT. By default, all PLCs communicate with data historian to report operational process data. PbNN retrieves the data from the historian to avoid the operational overheads in the PLC network. Each module used in the offline phase (Section IV) is described next.

TABLE III
INVARIANTS FOR FLOW AND LEVEL SENSORS IN STAGES P1-P3

Invaraint	Independent state variable	Dependent state variable
I1	FIT101(t)	FIT101(t-t _k), MV101(t-t _k)
I2	LIT101(t)	LIT101(t-t _k), FIT101(t-t _k), FIT201(t-t _k)
I3		LIT101(t-t _k), MV101(t-t _k), P601(t-t _k), P101(t-t _k)
I4	FIT201(t)	FIT201(t-t _k), MV201(t-t _k), P101(t-t _k)
I5	LIT301(t)	LIT301(t-t _k), FIT201(t-t _k), FIT301(t-t _k)
I6		LIT301(t-t _k), MV201(t-t _k), P301(t-t _k)

TABLE IV
PARAMETER SETTING

S.No	Hyperparameter	Range
1	Filters in convolution layer	[32,64,128]
2	Hidden units in fully connected layer	[5-100]
3	t_k	[2-10]
4	Number of epochs	[10,20,50]
5	Batch size	[32,64]

1) *Data Preprocessor Module*: In SWaT, each sensor measurement and the status of each actuator is stored in the historian once every second. For experimentation, we used the records from the historian while the plant is operating in a steady state. Furthermore, records extracted from the historian were converted to a format compatible with the input requirement of the anomaly detector under test. As each record needs to be represented in a numeric format, the status of an actuator is converted to “2” in case of “OPEN” or “ON,” and “1” in case of “CLOSE” or “OFF.” The transient state of the actuators is also represented as “1.”

2) *Component Set Extractor*: In this experiment, we monitor the behavior of flow rate and level sensors in stages P1-P3, namely, FIT101, LIT101, FIT201, LIT301, and FIT301. The dependency of each sensor with other SWaT components is obtained from the invariants (Table III) through the “component set extractor.” It can be noticed that we have considered all possible invariants holding the relationships across the discrete and continuous state variables to estimate the measurements from the above-mentioned sensors. Several models representing these relationships are generated by the “model generator” and the best model for each invariant is identified by the “model pruner.”

3) *Model Generator and Model Pruner*: The historical SWaT dataset representing its normal operation is divided into a ratio of 60:40 for training and validation purposes. The training dataset was utilized for the identification of the best DCNN architecture that represents each invariant. During the training process, three distinct CNN architecture with varying sets of convolution and pooling layers are, namely: 1) DCNN₁ (CNN with one set of convolution and pooling layer); 2) DCNN₂ (CNN with two sets of convolution and pooling layers); and 3) DCNN₃ (CNN with three sets of convolution and pooling layers). Table IV presents the parameter settings of the considered CNN architecture for the proposed anomaly detector. In addition to the hyperparameters of CNN (refer Section IV-B), we have also considered another parameter named time delay (t_k). As the experiments were conducted on the operational plant, the change in discrete state variables

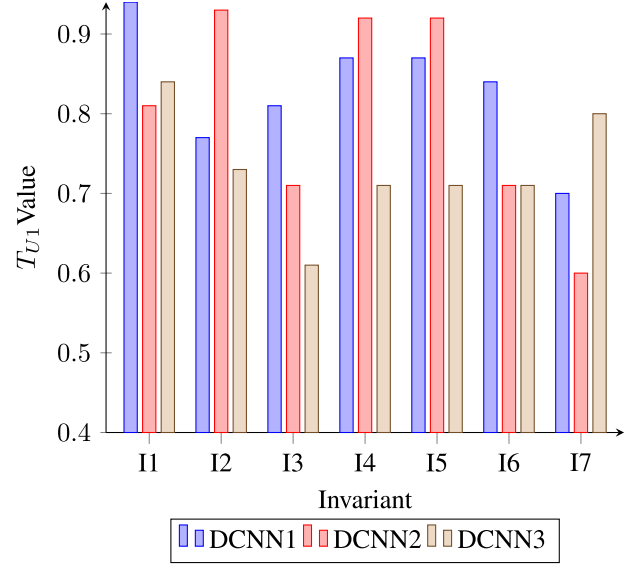


Fig. 4. T_{u1} computed using (5) for various CNN architectures.

has a cascading effect in the values of the continuous state variables. As discussed in Section IV-A, (3) represents a non-linear relationship among the state variables with single-order time delay. However, in real time, the opening or closing of a valve does not happen instantaneously upon the receipt of a control command. Instead, there exists a transient state of a few seconds duration such that we experience an approximately 11% increase in the inflow rate of water, i.e., as measured by the corresponding FIT. Although we neglect the existence of the transient state in this study, the time delay needs to be accounted for during model creation to minimize the false positives. Therefore, (3) has been modified as follows:

$$x_t(t+1) = f(x_1(t), x_2(t-t_k), x_3(t-t_k)) \quad (8)$$

where t_k corresponds to the time delay in moving from one stable to another stable state. As shown in Fig. 4, the best architecture with fine tuned hyperparameters is obtained through the T_{u1} computed from (5). Furthermore, in order to minimize the chance of overfitting during the training process, early stopping criteria with 10% of training dataset were utilized.

4) *Anomaly Scorer*: After building the independent models with fine-tuned hyperparameters for the considered sensors, the validation dataset is utilized to compute the value of several parameters discussed in Section IV-C. Furthermore, the dataset used for the validation process represents the behavior

of the plant in late 2015. Over time, i.e., there have been several improvements and maintenance works carried out on the plant such as, for example, replacement of faulty components, changes in the control codes of PLCs, and the installation of new sensors and actuators. Before analyzing the performance of the proposed anomaly detector, it is important to ensure the reliability of the computed statistical parameters. Hence, the plant was operated for 2 h under normal state to fine-tune the statistical parameters with the current behavior.

During the real-time implementation of PbNN in SWaT, there might be a possibility of experiencing a few short-term deviations due to noise and irregularity in the change of sensor values or actuator state. Such deviations should not be treated as anomalies since these are not the outcomes of a cyber attack. Two parameters are introduced to take such deviations into account and minimize false positives. These parameters are time window (T_w), and window size (S_w). Generally, T_w determines the allowable number of times the deviations ($r(t) > \text{UCL}$ or $r(t) < \text{LCL}$) can occur in a window size of S_w . Thus, the alert ($A(t)$) at any point in time t is defined in (10) as

$$f(t) = \begin{cases} 1, & (r(t) > \text{UCL} \text{ or } r(t) < \text{LCL}) \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

$$A_t = \begin{cases} \text{Anomaly,} & \left(\prod_{t=1}^{S_w} f(t) \right) > T_w \\ \text{Normal,} & \text{otherwise.} \end{cases} \quad (10)$$

B. Performance Metrics

Generally, the performance of an anomaly detector for ICS is evaluated in terms of their attack detection ability. Commonly used quality metrics that quantify the detection capability are detection rate (11) and false alarm rate (12). Additionally, we have considered a metric referred to as a conflict index factor (13) that determines tradeoff between the two conflict parameters, i.e., detection rate and false alarm rate. As PbNN is a combination of design and data-driven approaches, several machine learning-based anomaly detectors, such as CNN, MLP neural network [11], probabilistic neural network (PNN) [21], and DAE [10] along with the design centric approach, namely, DAD [7] designed for SWaT have been used for performance comparison

$$\text{Detection rate } (D_r) = \frac{T_p}{T_p + F_n} \quad (11)$$

$$\text{False alarm rate } (F_r) = \frac{F_p}{F_p + T_n} \quad (12)$$

$$\text{Conflict index Factor } (CiF) = (c_1 * D_r) - (c_2 * F_r) \quad (13)$$

where T_p is the number of samples that are correctly detected as “anomalous,” T_n is the number of samples that are correctly identified as “normal,” F_p is the number of samples that are incorrectly detected as “anomalous,” F_n is the number of samples that are misclassified as “normal,” and c_1 and c_2 are the user defined weights such that $c_1 + c_2 = 1$.

Anomaly detectors, such as DAD, PNN, MLP, and DAE, are designed in iTrust and are operational in SWaT. The CNN-based anomaly detector is implemented by the authors using

TABLE V
PERFORMANCE OF ANOMALY DETECTORS UNDER NORMAL OPERATION OF SWaT

S.No	Detectors	False positive rate (%)	
		Automated mode	Manual mode
1	PNN[21]	1.32	1.7
2	CNN	0	0.54
3	MLP[11]	0	0.5
4	DAE[10]	0	0
5	DAD[7]	0	0
6	PbNN	0	0

the Keras library with a default set of parameters. Furthermore, to have a fair comparison, the CNN model is trained with the same dataset utilized while training models in PbNN.

C. Results and Discussion

The test phase of the proposed anomaly detector was carried out in a live plant under two scenarios. In one scenario, the plant was run without any attacks, i.e., under normal conditions, to compare the performance of anomaly detectors against false positives. In another scenario, attacks listed in Tables I and II were launched against the plant as described in Section V-B. Each attack was launched, and remained active, until its intended objective was realized. Importantly, to avoid physical damage to the components, such as pumps and valves, few alterations were made to the water level markers of tank T101 and T301 such that there is no physical overflow in the affected tanks and no tanks are fully empty. In the following discussion, we claim that an attack is considered detected only when an alert is raised by the corresponding model designed for a particular state variable. For example, during the launch of attack A3, when the model monitoring the behavior of FIT101 raised an alert, we consider the attack as detected. In the case of any other model raising an alert, we declare it as a false alarm. Furthermore, we launched two kinds of stealthy attacks, namely, “surge” and “bias” [19]. In a surge attack, the attacker attempts to realize maximum damage to the physical components as soon as possible. In a bias attack, the attacker injects a small value over a longer period of time. However, in both cases, the magnitude in the difference between the actual and the mutated measurements is kept small.

D. Performance of Anomaly Detectors During SWaT in Normal Operation

To demonstrate the effectiveness of the anomaly detectors in terms of false positives, SWaT was operated continuously for 2 h under normal conditions. During the first hour, SWaT was in an automated mode and then shifted to the manual mode where the water storage capacity of the UF feed tank (T301) is increased. Generally, an ICS operates in a dynamic environment with several modes of operations based on the domain constraints. For example, the water treatment plant may operate in a different mode during a holiday season than otherwise.

An anomaly detector designed for ICS should not raise alarms during the behavioral change incurred during the mode shift. As shown in Table V, the anomaly detectors with

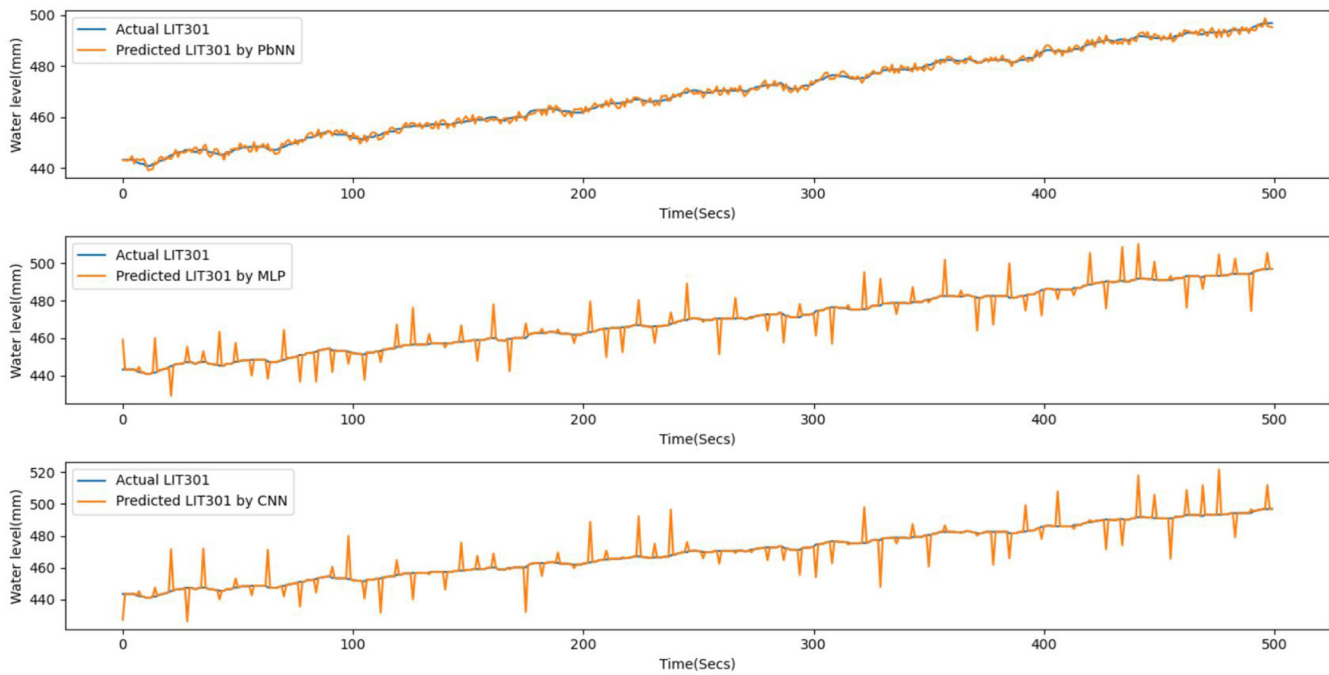


Fig. 5. Performance of PbNN, MLP, and CNN-based anomaly detectors during SWaT operation in manual mode.

design knowledge show better performance by achieving a 0% false-positive rate compared with data-centric approaches in the manual mode operation of SWaT. From Fig. 5, the spikes in the behavior of MLP and CNN-based anomaly detectors represent the false alarms generated during the SWaT operation in the manual mode. Such unexpected spikes are due to the nature of the training dataset. The SWaT-2015 dataset available in [16] consists of a LIT301 level sensor for tank T301 operated between 800 and 100 mm. During the manual mode, the range was altered to 250–1000 mm. Such alteration cannot be considered as anomalous since the relationship between LIT301 and its correlated state variables, e.g., FIT201, FIT301, MV201, and P301, were not altered. This scenario is one of its kind wherein several such exist in a large-scale plant.

The results reported in Table V emphasize the need for design knowledge while building an anomaly detector for ICS that adheres to the dynamic operating conditions of ICS. Another notable reason underlying the 0% false-positive rate for PbNN is due to the impact of careful tuning of the parameters discussed in Section IV-C.

E. Performance of Anomaly Detectors During SWaT in Attack Scenarios

Unlike the previous scenario, SWaT was operated in an automated mode with the default configuration and parameter settings during attack launch. Cyber attacks are launched against the operational SWaT via level 1, i.e., from the SCADA workstation. As discussed in Section V-B, each attack is launched five times at different steady states of SWaT with increasing values of X and Y as in Tables I and II. The main objective of this study is to design an efficient anomaly

detector for ICS that possess a high detection rate and low rate of false alarms. Thus, the effectiveness of the data-driven anomaly detectors with varied design knowledge is determined by keeping the performance of DAD as a baseline. DAD is a fully functional design-centric anomaly detector in SWaT. It makes use of the fundamental laws of physics and chemistry to create deterministic rules manually that govern the plant operation at all times. Importantly, DAD has proven its effectiveness in several case studies reported in [7], [20], [22], and [23].

As indicated in Table VI, all single (A1–A12) and multipoint (A13–16) attacks launched against the SWaT components were detected by DAD and PbNN before the intention was realized. However, other detectors, such as PNN, MLP, CNN, and DAE, do not exhibit similar performance. Of the 60 instances of single-point attacks, PNN and DAE detected, respectively, 16, and 42 anomalies resulting from the attacks. Similarly, for multipoint attacks, of the 20 instances launched, PNN detected four and DAE detected 19. Furthermore, MLP and CNN-based anomaly detectors exhibit similar performance in detecting 55 single point attacks out of 60, and 19 multipoint attacks out of 20. From Table VI, it can be observed that anomaly detectors except PbNN and DAD detected fewer anomalies after the attack was removed. This is due to the nature of the attack launched. As discussed in Section III-B, to avoid suspicion on the operator's side, the difference in the magnitude of the actual and the altered measurement was kept small.

Table VII compares the performance of each detector against DAD using the metrics defined in (11)–(13). As our study is confined to the detection of anomalies that occur due to cyber attacks in the physical process, considered as a rare event in this work, the values of c_1 and c_2 in (13) are

TABLE VI
EFFECTIVENESS OF ANOMALY DETECTORS

Attack Type	Attack ID	Attacks Launched	Attacks detected					
			PNN[21]	MLP[11]	CNN	DAE[10]	DAD[7]	PbNN
Single-point attack	A1	5	1	4	4	3	5	5
	A2	5	2*	5*	5*	4*	5	5
	A3	5	2	5	5	3	5	5
	A4	5	2*	4*	4	3	5	5
	A5	5	1	4	4	4*	5	5
	A6	5	1	5*	5*	4	5	5
	A7	5	2*	5	5	3	5	5
	A8	5	1*	5	4	3	5	5
	A9	5	0	4	4	4*	5	5
	A10	5	1*	4	5*	4	5	5
	A11	5	1	5*	5	3	5	5
	A12	5	2	5	5*	4	5	5
	Total	60	16	55	55	42	60	60
Multi-point attack	A13	5	2	5	5	5	5	5
	A14	5	1	5*	5	5*	5	5
	A15	5	1*	5	4*	4	5	5
	A16	5	0	5	5	5	5	5
	Total	20	4	20	19	19	20	20

*Detected after the attack was removed in few cases

TABLE VII
EFFECTIVENESS OF ANOMALY DETECTORS IN TERMS OF
PERFORMANCE METRICS DEFINED IN (11)–(13)

Anomaly detector	Performance metric		
	Dr (%)	Fr (%)	CiF
PNN[21]	25	0.6	9.64
CNN	93.75	0.05	37.47
MLP[11]	92.5	0.06	36.96
DAE[10]	76.25	0.19	30.38
DAD[7]	100	0	40
PbNN			

0.4 and 0.6, respectively, thus assigning a higher weight to the rate of false alarms. Analyzing the overall performance of the detectors, we conclude that PbNN dominates other techniques and achieves performance similar to that of DAD. Furthermore, the increase in the detection rate of 5% between CNN and PbNN indicates the impact of design knowledge in the data-centric approach in the detection of process anomalies. Although PbNN exhibits performance similar to that of DAD, the average attack detection time of PbNN (6.7 s) is slightly higher compared to that of DAD (4.1 s). This is due to the effect of parameters, such as T_w and S_w , which were introduced to minimize the false alarms due to temporal glitches and noise. Nevertheless, PbNN detects anomalies due to cyber attack before the attack intention is achieved.

F. Need for PbNN in the Presence of DAD

Designing a CI is a complex task since it integrates knowledge from several domains in science and engineering. Basically, “design knowledge” of a plant refers to its architecture, components with their specifications, and other computing infrastructure details. Such design knowledge can be obtained from the operator manuals, design documents, and vendor specification manuals. To ensure the normal plant operation and for maintenance and upgrade purposes, design knowledge is found to be mandatory. DAD [7] is one such design-based anomaly detector where the invariants are

created manually for SWaT. It requires a deep understanding of the plant design for its better performance. For a small testbed such as SWaT, the absence of an automated tool for the design of an anomaly detector is acceptable. However, for commercial plants deployed across cities, the existence of multiple heterogeneous components, their complex interactions, and various process specifications demands an automated tool for the design of an effective anomaly detector.

Thus, the data-centric approach based on machine learning algorithms and computational intelligent techniques appears to be the best alternative for a design-centric approach. However, the commercially available solutions based on a data-centric approach discussed in Section II do not meet the requirements of the anomaly detector for ICS reported by Ahmed *et al.* [24]. Thus, we attempt to use a data-centric approach to create an anomaly detector along with a significantly less understanding of the plant design compared with that required in the design-centric approach. Furthermore, the approach used must be computationally attractive and satisfy the real-time anomaly detection requirements for ICS. As a result, we propose PbNN, which takes the plant’s P&ID diagram (design knowledge) and operational data as an input to model the higher order and nonlinear relationships among the correlated components during their normal operation. Since the experimental results are encouraging, the methodology of PbNN seems promising for plants much larger than SWaT.

G. Research Questions and Contributions

The research questions formulated in Section I are revisited in the context of the experimental results presented above.

RQ 1: How accurately PbNN learns process dynamics of the underlying ICS through the application of design knowledge and operational data?

The subprocesses of the underlying ICS obey the laws of physics [19]. A formal physical model can be derived from these laws along with the availability of the operational data. For the considered water treatment plant, the dependencies

across components are captured in the form of invariants through the application of simple laws of physics [7]. In this work, we attempt to capture such relationships using DCNN through the application of the historical data available in [16] and from the operational SWaT testbed. Furthermore, the performance of PbNN was improved by fine-tuning the values of the hyperparameters based on the current system behavior to effectively capture the higher order and nonlinear relationships among the state variables resulting in the prediction of the expected system behavior with minimal error. Observations from Fig. 4 reveal the significance of the PbNN in efficiently analyzing and predicting the behavior of SWaT components.

RQ 2: How effective is PbNN in detecting process anomalies compared with the existing state-of-the-art machine learning-based anomaly detectors?

From Table VI, we can observe a better performance of PbNN compared with other anomaly detectors in terms of the quality metrics considered. Furthermore, throughout this experiment, the overall attack detection rate of PbNN was found to be 100% and false alarm rate was 0%, which is not achieved by any other detector derived using machine learning. The use of CUSUM with PbNN proved to be an effective solution for detecting stealthy attacks in SWaT due to following reasons.

- 1) The CUSUM approach tracks the rate of change of residual errors regardless of how small the errors are.
- 2) The parameters, i.e., target, b , UCL, and LCL discussed in Section IV-C, were fine-tuned based on the normal behavior of SWaT using both historical and live data.
- 3) The extrinsic parameters T_w and S_w (Section VI-A4) were empirically computed to minimize the false alarms due to sensor noise and temporal glitches.

RQ 3: Does PbNN support the plant operator in the identification of attack target(s)?

Anomaly detectors designed for ICS are either univariate or multivariate. In the former case, a representative model is built by using the temporal dependency of the single sensor or actuator, while in the latter case the process models are derived from laws of physics, e.g., using the interactions among the components. PbNN is a multivariate detector. Although the design of univariate anomaly detectors is relatively easy and computationally attractive, they fail to detect coordinated attacks. The key reason for such failure is that univariate detectors are based on the behavior of single components. However, in the case of PbNN, a multivariate anomaly detector has the ability to effectively detect multipoint coordinated attacks as is evident from the discussion in Section VI. In addition, PbNN enables a plant operator to visualize the current and predicted behavior of each component in real time and thus identify the attack targets.

VII. CONCLUSION

ICSs are omnipresent and increasingly deployed in CI. A skilled attacker exploits several vulnerabilities in these systems and successfully launches their sophisticated, large-scale cyber

attacks. In practice, the design of defense mechanisms against these attacks is a challenging task. However, deep inspection of the nature and behavior of several attacks against ICS led to the conclusion that they all influence the physical behavior of components. This observation has led to PbNN for real-time monitoring of process dynamics of the system.

In contrast to the existing approaches, PbNN benefits from the merits of both design and data-centric approaches for the detection of process anomalies. In PbNN, the interactions among the physical devices are captured through design knowledge and modeled using DCNN. Furthermore, the statistical CUSUM approach is integrated with DCNN for improved decision making. The approach was evaluated in a water treatment plant named SWaT and compared against several single and multipoint coordinated attacks. The experimental analysis reveals that PbNN exhibits a high level of detection accuracy with zero false alarms compared with existing state-of-the-art anomaly detectors. The outcome of the study reported here is encouraging and enables further experimentation where PbNN can be applied to a larger set of state variables. Last but not the least, PbNN performs nearly as close to a pure design-centric detector, namely, DAD, thus revealing the advantages of using a design-flavored data-centric approach over a pure design centric approach for the creation of anomaly detectors for physical processes.

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